

Disproof of conjecture 00000001857:
the closed form gives $c_3 = \frac{2-\sqrt{3}}{2} \cdot \frac{1+\sqrt{5}}{2} \approx 0.2168 < 1$,
so $\tau(G) \not\sim c_3^n$ and $c_3 \neq 1.175\dots$

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1 The conjecture and the clauses refuted

Statement (English, verbatim up to typesetting). Definition: The spanning-tree count $\tau(G)$ of a random $d = 3$ regular graph. Conjecture: $\tau(G)$ is asymptotically c_3^n (an exponential constant) with

$$c_3 = \left(\frac{\sqrt{3}-1}{2}\right)^2 \exp\left(\int \log(3-2\cos\theta) d\theta / 4\pi\right)$$

(a McKay-type formula); the numerical limit is $1.175\dots$, and the closed form is computable. The Chinese line states the same: the asymptotics of $\tau(G)$ is c_3^n with the same formula (called a “McKay-norm type formula”), the numerical limit is $1.175\dots$, and an explicit closed form is computable.

Reading. n is the number of vertices; a random 3-regular graph on n vertices (n even) is a uniformly chosen 3-regular simple graph on the labelled vertex set $\{0, \dots, n-1\}$. The integral is over a full period $[0, 2\pi]$ ($[-\pi, \pi]$ gives the same value by periodicity), and “ $/4\pi$ ” divides the integral by 4π . Our main bound holds for every window $[a, b]$ with $0 \leq b-a \leq 4\pi$, so it also covers $[-\pi, \pi]$, $[0, \pi]$ and $[0, 4\pi]$. “The numerical limit is $1.175\dots$ ” means a real number whose decimal expansion begins 1.175 , i.e. a number in $[1.175, 1.176)$. For “ $\tau(G)$ is asymptotically c_3^n ” we treat three readings: (A) $\tau(G_n)/c_3^n \rightarrow 1$; (B) $\tau(G_n) = O(c_3^n)$ (hence also Θ); (C) the exponential-rate reading $\tau(G_n)^{1/n} \rightarrow c_3$.

What is refuted. (1) The conjunction “ c_3 equals the stated formula and its numerical value is $1.175\dots$ ”: the formula is < 1 for every window of length $\leq 4\pi$ and equals ≈ 0.2168 on $[0, 2\pi]$. (2) The main asymptotic clause with c_3 given by the formula: under each of (A), (B), (C) it fails for *every* sequence of connected finite graphs with $n \rightarrow \infty$, because $\tau \geq 1$ while $c_3^n \rightarrow 0$; random 3-regular graphs are connected asymptotically almost surely, so it fails for them. Both clauses are main-clause content; we do not use the parentheticals or the word “type”. Not refuted: a reading that discards the formula and keeps only “ $\tau(G) \approx 1.175^n$ ”.

2 Definitions (as formalized)

Definition 1 (spanning-tree count). For a simple graph G on a vertex type V , `tau G` is the number of simple graphs T on V with $T \leq G$ (every edge of T is an edge of G) and T a tree (Mathlib `SimpleGraph.IsTree`: connected, hence nonempty, and acyclic). These are exactly the spanning trees of G . Lean theorem `tau_eq_card_edgeSets` proves that the same number counts the edge subsets $s \subseteq E(G)$ whose spanning graph `fromEdgeSet s` is a tree. (Lean uses `Nat.card`; every theorem below assumes V finite.)

Definition 2 (closed form). $c_3 = ((\sqrt{3} - 1)/2)^2 \cdot \exp((\int_a^b \log(3 - 2 \cos \theta) d\theta)/(4\pi))$ and $c_3 = c_3 = c_3(0, 2\pi)$. Here $\log$ is the real natural logarithm and $3 - 2 \cos \theta \geq 1 > 0$, so no branch issue arises.

Definition 3 (uniform random 3-regular graph). $\text{reg3 } n$ is the finite set of simple graphs on $\text{Fin } n$ that are 3-regular (`Mathlib.IsRegularOfDegree 3`); $\text{prob } n P = |\{G \in \text{reg3 } n : P(G)\}|/|\text{reg3 } n|$.

3 Proof

Lemma 4 (`one_le_tau`). If V is finite and G is connected then $\tau(G) \geq 1$.

Proof. `Mathlib.SimpleGraph.Connected.exists_isTree_le` gives a tree $T \leq G$; the set of spanning trees is nonempty and finite. $\square$

Theorem 5 (`c3n_lt_one`, `c3n_not_numerical`, `c3_lt_one`). If $a \leq b \leq a + 4\pi$ then $0 < c_3(a, b) < 1$; in particular $c_3(a, b) \notin [1.175, 1.176]$. This applies to $[0, 2\pi]$ and $[-\pi, \pi]$.

Proof. Since $-1 \leq \cos \theta \leq 1$ we have $1 \leq 3 - 2 \cos \theta \leq 5$, so the integrand is continuous and $\int_a^b \log(3 - 2 \cos \theta) d\theta \leq (b - a) \log 5 \leq 4\pi \log 5$. Hence the exponential is at most 5. Also $((\sqrt{3} - 1)/2)^2 = (2 - \sqrt{3})/2$ and $\sqrt{3} > 1.732$, so the product is at most $5(2 - \sqrt{3})/2 < 0.67 < 1$. $\square$

Theorem 6 (`integral_eq`, `c3_eq`, `c3_bounds`). $\int_0^{2\pi} \log(3 - 2 \cos \theta) d\theta = 2\pi \log q$ with $q = (3 + \sqrt{5})/2$, hence $c_3 = \frac{2-\sqrt{3}}{2} \cdot \frac{1+\sqrt{5}}{2}$ and $0.2167 < c_3 < 0.2169$.

Proof. Since $q^2 = 3q - 1$, $|e^{i\theta} - q|^2 = 1 - 2q \cos \theta + q^2 = q(3 - 2 \cos \theta)$, so $\log |e^{i\theta} - q| = \frac{1}{2}(\log(3 - 2 \cos \theta) + \log q)$. `Mathlib.circleAverage_log_norm_sub_const_2` states that for $|q| > 1$ the average of $\log |z - q|$ over the unit circle is $\log |q|$. Averaging the identity gives $\frac{1}{2\pi} \int_0^{2\pi} \log(3 - 2 \cos \theta) d\theta = \log q$. Then $\exp(2\pi \log q/4\pi) = \sqrt{q} = (1 + \sqrt{5})/2$, because $((1 + \sqrt{5})/2)^2 = q$. The enclosure follows from $1.7320 < \sqrt{3} < 1.7321$ and $2.2360 < \sqrt{5} < 2.2361$. $\square$

Theorem 7 (`tau_ratio_tendsto_atTop`, `not_tau_asymp`, `not_tau_isBig0`, `not_tau_rpow_tendsto`, `conjecture1857_false`). Let (G_k) be connected graphs on finite vertex types V_k with $n_k = |V_k| \rightarrow \infty$, and let $c = c_3(a, b)$ with $a \leq b \leq a + 4\pi$. Then $\tau(G_k)/c^{n_k} \rightarrow \infty$ (so it does not tend to 1), $\tau(G_k)$ is not $O(c^{n_k})$, and $\tau(G_k)^{1/n_k} \not\rightarrow c$.

Proof. $0 < c < 1$, so $c^{n_k} \rightarrow 0^+$ and $1/c^{n_k} \rightarrow \infty$; and $\tau(G_k)/c^{n_k} \geq 1/c^{n_k}$ by the lemma. If $\tau(G_k) = O(c^{n_k})$ then $\tau(G_k) \rightarrow 0$, contradicting $\tau \geq 1$. Finally $\tau(G_k)^{1/n_k} \geq 1$ for all k , so any limit is $\geq 1 > c$. $\square$

Theorem 8 (`prob_good_le_prob_disconnected`, `not_prob_good_tendsto_one`). Let $c = c_3(a, b)$ with $a \leq b \leq a + 4\pi$ and let $\varepsilon \leq 1$. There is N such that for all $n \geq N$, $\text{prob } n (|\tau(G)/c^n - 1| < \varepsilon) \leq \text{prob } n (G \text{ disconnected})$. Consequently, if $\text{prob } (2m) (G \text{ disconnected}) \rightarrow 0$, then $\text{prob } (2m) (|\tau(G)/c^{2m} - 1| < \varepsilon) \rightarrow 0$, and in particular it does not tend to 1.

Proof. Choose N with $c^n < 1/2$ for $n \geq N$. A connected G then has $\tau(G)/c^n \geq 2$, so $|\tau(G)/c^n - 1| \geq 1 \geq \varepsilon$. Thus the good event lies inside the disconnection event, and the probabilities compare. The second part is a squeeze argument. $\square$

The hypothesis of the second part is a known theorem, not proved in Lean: Wikipedia's article on random regular graphs states that "a random r -regular graph of large size is asymptotically almost surely r -connected" for $r \geq 3$. In particular a random 3-regular graph is connected a.a.s. Hence $\tau(G)/c_3^n \rightarrow \infty$ in probability, and $\tau(G) \sim c_3^n$ fails in probability and a.a.s. The deterministic theorem above needs no probabilistic input: it holds along every sequence of connected graphs.

4 Lean formalization and axiom audit

File `lean/Conjecture1857/Basic.lean` (315 lines, Lean 4.33.1, Mathlib v4.33.1; namespace `C1857`; only import `Mathlib`). Its main theorem is

```
theorem conjecture1857_false :
  c3 ∉ Set.Ico (1.175 : ℝ) 1.176 ∧
  ∀ {V : ℕ → Type} [∀ k, Finite (V k)] (G : ∀ k, SimpleGraph (V k)),
    Tendsto (fun k => Nat.card (V k)) atTop atTop → (∀ k, (G k).Connected) →
      ¬ Tendsto (fun k => (tau (G k) : ℝ) / c3 ^ Nat.card (V k)) atTop (ℕ 1)
```

(here $\mathcal{N}$ is Mathlib’s neighbourhood filter `nhds`), together with `c30n_lt_one`, `c30n_not_numerical`, `c3_lt_one`, `integral_eq`, `c3_eq`, `c3_bounds`, `not_tau_asymp`, `not_tau_isBig0`, `not_tau_rpow_tendsto`, `tau_eq_card_edgeSets`, `prob_good_le_prob_disconnected` and `not_prob_good_tendsto_one`. The last one takes a.a.s. connectivity as an explicit hypothesis `hconn`, along the even vertex counts $n = 2m$. It adds no axiom. `lake build` succeeds, and the file contains no `sorry` or `native_decide`. For each of the theorems listed, `#print axioms` reports exactly [`propxt`, `Classical.choice`, `Quot.sound`].

5 Relation to the earlier submission

PR #236 (orionsheep, closed) computed the same value, ≈ 0.2168 , but did so only in prose. The reviewer wrote: “the main theorem is a vacuous implication over an abstract `Nat` (any `c2` $< 1000 \neq 2350$, plus $9 > 4 \wedge 25 > 20$); none of $\tau(G)$, c_3 , the integral, `exp`, or $\sqrt{}$ appears in Lean, and the entire closed-form evaluation ≈ 0.217 vs 1.175 exists only in prose.” This submission formalizes the actual objects. These are the spanning-tree count $\tau(G)$ of a Mathlib simple graph, the closed form with the real integral, `exp` and $\sqrt{}$, the exact evaluation of that integral, and the uniform random 3-regular graph on `Fin n`. It also refutes the asymptotic clause itself, for every sequence of connected graphs, which the earlier submission did not address. The earlier submission also said the literature constant is ≈ 1.216 ; we do not rely on any literature constant.

Sources

- Wikipedia, “Random regular graph”, https://en.wikipedia.org/wiki/Random_regular_graph (retrieved 2026-10-05): “a random r -regular graph of large size is asymptotically almost surely r -connected”.
- PR #236 and its review, <https://github.com/The-Last-Math-Competition/The-Last-Math-Competition/pull/236>.
- Mathlib: `SimpleGraph.Connected.exists_isTree_le` (Combinatorics/SimpleGraph/Acyclic.lean), `circleAverage_log_norm_sub_const_2` (Analysis/SpecialFunctions/Integrals/PosLogEqCircleAverage.lean).